Title

Exploring Customer Segmentation Through Hierarchical Clustering

Introduction

Clustering is a technique in unsupervised machine learning used to group similar data points together so that patterns and hidden relationships can be identified. One of the most widely used approaches is **Hierarchical Clustering**, which organizes data into a tree-like structure (dendrogram). Unlike k-means, hierarchical clustering does not require the number of clusters to be specified in advance, making it useful for exploratory data analysis.

In this experiment, hierarchical clustering was applied on a customer dataset to identify segments based on **Age, Annual Income, Spending Score, and Gender**. Both unscaled and scaled versions of the data were analyzed, and dendrograms as well as cluster visualizations were used to interpret the results.

Required Libraries

- Pandas
- Matplotlib
- Scikit-learn
- Scipy

Theory

Clustering groups similar data points such that intra-cluster similarity is maximized and inter-cluster similarity is minimized.

Two main approaches in hierarchical clustering:

1. **Top-Down (Divisive)** – start with one cluster and recursively split it.
2. **Bottom-Up (Agglomerative)** – start with each point as its own cluster and merge step by step.

Code and Explanation

Code	Explanation

`import pandas as pd`
`import matplotlib.pyplot as plt`
`from sklearn.preprocessing import LabelEncoder, StandardScaler`
`from sklearn.cluster import AgglomerativeClustering`
`from scipy.cluster.hierarchy import dendrogram, linkage`
 	
`df = pd.read_csv('/content/drive/MyDrive/Colab Notebooks/PRL_12_8_2025.csv')`
	Libraries for clustering, visualization, and preprocessing are imported. Dataset is loaded into a pandas DataFrame.

`fig, axes = plt.subplots(1, 3, figsize=(18, 5))`
`axes[0].scatter(df['Annual Income (k$)'], df['Spending Score (1-100)'])`
`axes[1].scatter(df['Age'], df['Spending Score (1-100)'])`
`axes[2].scatter(df['Age'], df['Annual Income (k$)'])`
`plt.show()`
	Three scatter plots were created: (1) Income vs Spending, (2) Age vs Spending, (3) Age vs Income. This helps in understanding patterns between features.

`le = LabelEncoder()` `df['Gender_Encoded'] = le.fit_transform(df['Genre'])`	Label encoding is applied to convert categorical column *Genre* (Male/Female) into numeric form.

Code	Explanation

`X = df[['Annual Income (k$)',`
`  'Spending Score (1-100)',`
`  'Gender_Encoded']].values`
 	
`agg_clustering = AgglomerativeClustering(n_clusters=5,`
`  metric='euclidean',`
`  linkage='ward')`
 	
`df['Cluster'] = agg_clustering.fit_predict(X)`
	Features (Income, Spending, Gender) were used for Agglomerative Clustering with 5 clusters. Cluster labels were stored in new column *Cluster*.

`plt.figure(figsize=(12, 6))`
`linkage_matrix = linkage(X, method='ward')`
`dendrogram(linkage_matrix, truncate_mode='level', p=5)`
`plt.show()`
	A dendrogram was generated to visualize hierarchical merging of customers without scaling.

`plt.scatter(df['Annual Income (k$)'],`
`  df['Spending Score (1-100)'],`
`  c=df['Cluster'], cmap='viridis')`
`plt.show()`
	Scatter plot showing customer clusters (without scaling) with colors representing cluster assignments.

Code	Explanation

``python
scaler = StandardScaler()
X_scaled = scaler.fit_transform(
df[['Annual Income (k$)',
'Spending Score (1-100)',
'Gender_Encoded']])
agg_clustering_scaled = AgglomerativeClustering(
n_clusters=5, metric='euclidean', linkage='ward')
df['Cluster_Scaled'] = agg_clustering_scaled.fit_predict(X_scaled)
``
 	
	Data was standardized using StandardScaler, ensuring each feature contributes equally. Clustering was repeated on scaled data.

``python
plt.figure(figsize=(12, 6))
linkage_matrix_scaled = linkage(X_scaled, method='ward')
dendrogram(linkage_matrix_scaled, truncate_mode='level', p=5)
plt.show()
``
 	
	A dendrogram was generated using scaled data to compare clustering structures.

``python
plt.scatter(df['Annual Income (k$)'],
df['Spending Score (1-100)'],
c=df['Cluster_Scaled'], cmap='viridis')
plt.show()
``
 	
	Final scatter plot showing scaled clustering results. Clearer boundaries are visible after scaling.

Output

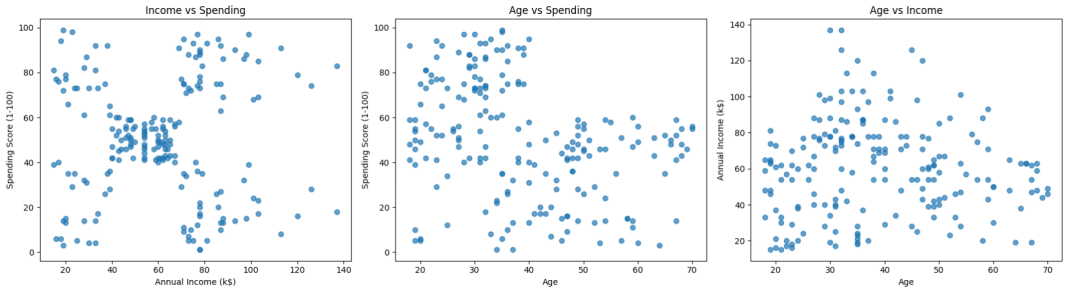

Figure 1: Scatter plots: Income vs Spending, Age vs Spending, Age vs Income

```
Gender encoding mapping: {'Female': np.int64(0), 'Male': np.int64(1)}
    Genre  Gender_Encoded
0    Male               1
1    Male               1
2  Female               0
3  Female               0
4  Female               0
```

Figure 2: Gender encoded values (Male=1, Female=0)

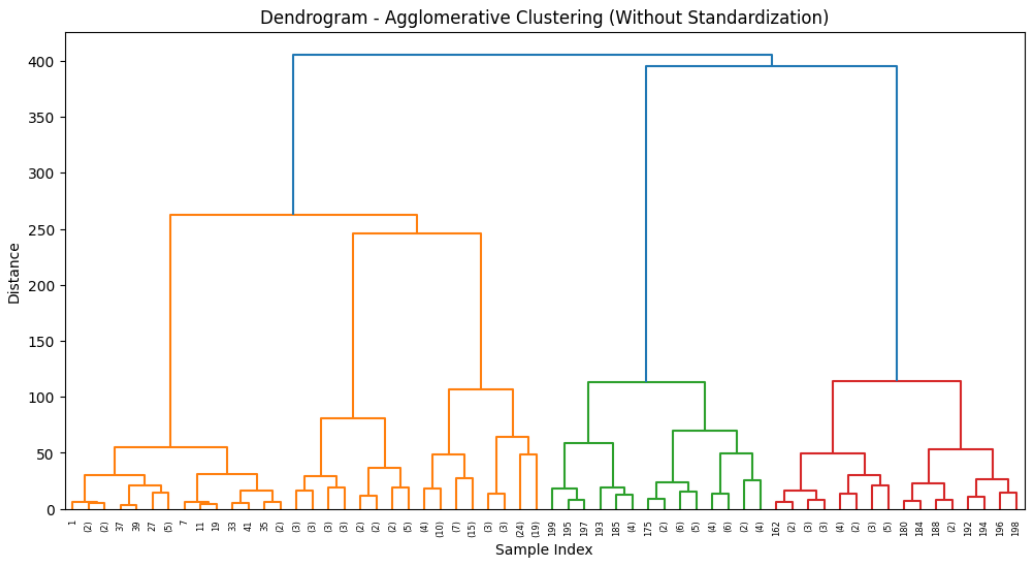

Figure 3: Dendrogram - Agglomerative Clustering (Unscaled Data)

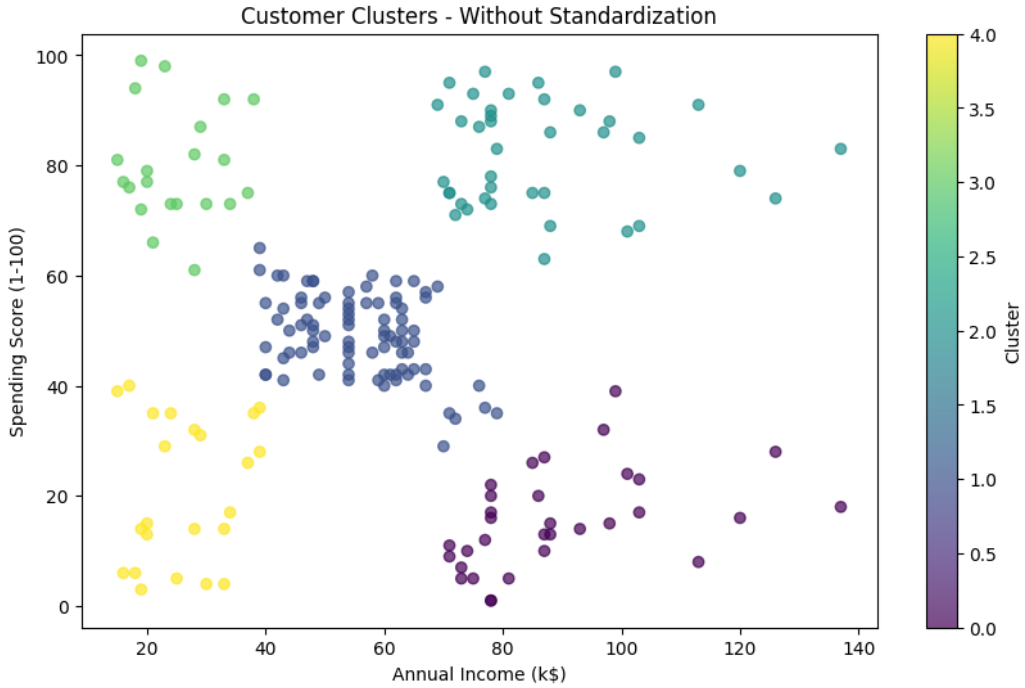

Figure 4: Clusters Visualization (Unscaled Data)

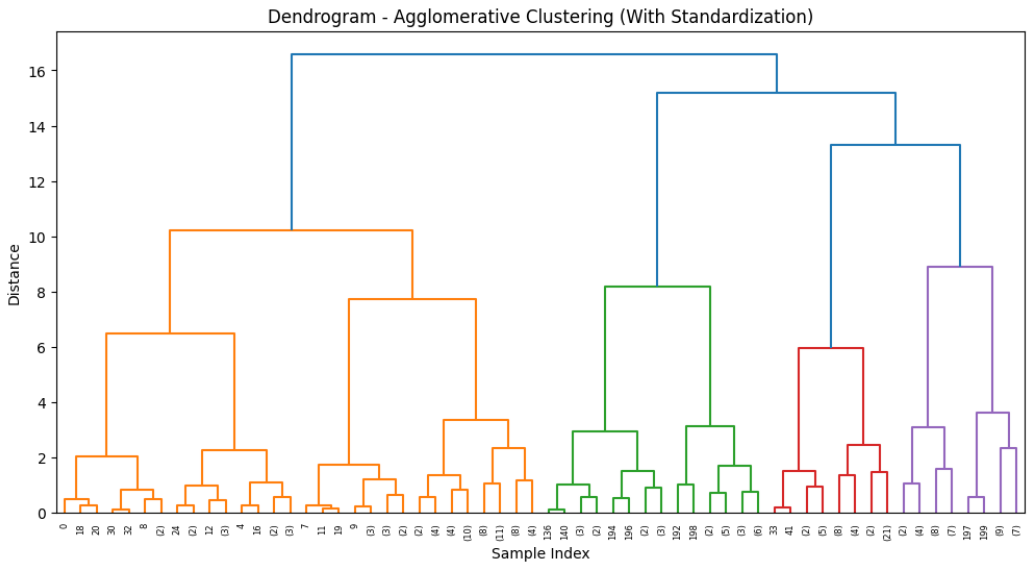

Figure 5: Dendrogram - Agglomerative Clustering (Scaled Data)

Figure 6: Clusters Visualization (Scaled Data)

Discussion

Hierarchical clustering was successfully applied to customer data including income, spending, age, and gender. Without standardization, features with larger ranges (such as income) had more influence on clustering. After applying StandardScaler, clusters became more balanced and meaningful, as all features contributed equally.

Dendrograms provided insight into possible numbers of clusters, while scatter plots gave a clear view of how customers were grouped. Including gender as a feature slightly adjusted the segmentation boundaries.

This experiment demonstrates how hierarchical clustering can be used for customer segmentation, helping businesses in targeted marketing and personalized recommendations.